

Short Papers

Biometric Identification through Hand Geometry Measurements

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Abstract—A work in defining and implementing a biometric system based on hand geometry identification is presented here. Hand features are extracted from a color photograph taken when the user has placed his hand on a platform designed for such a task. Different pattern recognition techniques have been tested to be used in classification and/or verification from Euclidean distance to neural networks. Experimental results, up to a 97 percent rate of success in classification, will show the possibility of using this system in medium/high security environments with full acceptance from all users.

Index Terms—Hand geometry, classification, verification, biometric systems, Euclidean distance, Hamming distance, Gaussian mixture models, radial basis functions.

1 INTRODUCTION

DAY by day, biometric identification is gaining more importance. Different techniques have been developed, each of them having its own advantages and disadvantages, according to user acceptance, cost, performance, etc. [4]. From all these techniques, hand geometry is considered to achieve medium security, but with several advantages compared to other techniques:

1. Medium cost as it only needs a platform and a low/medium resolution CCD camera.
2. It uses low-computational cost algorithms, which leads to fast results.
3. Low template size: from 9 to 25 bytes, which reduces the storage needs.
4. Very easy and attractive to users: leading to a nearly null user rejection.
5. Lack of relation to police, justice, and criminal records.

There are nearly no bibliographic references about hand geometry biometrics ([4]), although there is one commercial system available, developed by Recognition Systems Inc. ([8]). The hand geometry identification system detailed in this work has the typical architecture of any other biometric system [7]. It works in two phases: an enrollment one and a comparison one. In the enrollment phase, several photographs are taken from the user; these photographs are then preprocessed to enter the feature extraction block, where a set of measurements is performed. With the features extracted, the user's pattern is computed and stored in a central database or a portable storage media. In the verification phase, a single photograph is taken, preprocessed, and entered in

the feature extraction block. This single set of features is compared with the template previously stored, obtaining a ratio of likeness to determine who the user is whose hand has been photographed.

The comparison block can be configured in two ways: as a classifier (i.e., biometric recognition), where the features extracted are compared to all the users' templates to determine which is the user photographed, or as a verifier (i.e., biometric verification), where the user's identity is stated and, therefore, the sample is only compared with the pattern of the claimed user.

Throughout this work, an explanation of all the different blocks of the hand geometry biometric identification system designed will be given. Section 2 presents the feature extraction block, including preprocessing, measurements taken, and optimization of the template size through selection of representative features. In Section 3, the classification and verification problem will be studied. In Section 4, we present our experimental results, detailing the system used, the database, and final figures in classification and verification. The conclusions, according to the results obtained, and future research directions are presented in Section 5.

2 FEATURE EXTRACTION

The aim of this first block is to capture a sample of the user's biological data, process it, and extract a set of features that represent univocally that user among all the population that use the system. To achieve this, three tasks are performed: image capture, preprocessing, and features measurement. A large set of features are extracted, but from these only a subset have enough significance, i.e., are independent and unique for a determined user. To know which is the optimal subset for a determined error rate, statistical techniques, as detailed below, have been applied.

2.1 Image Capture

The sample signal is obtained with a CCD color camera, placed above a platform designed to guide the hand to a fixed location. Different views of the prototype designed can be seen in Fig. 1. The platform has six tops placed in determined positions to guide the placement of the hand. These tops are equipped with pressure sensors which, when all are activated, trigger the camera and the photograph is taken. The image captured is a 640 x 480 pixels color photograph in JPEG format. This image contains not only a view of the palm, but also the lateral view of the hand due to the placement of a mirror in the right side of the platform in order to be able to measure heights.

2.2 Preprocessing

After the image is captured, preprocessing is performed. The first step in the preprocessing block is to transform the color image into a black and white one where the background is eliminated. To achieve this, an arithmetic operation is performed with the different channels:

$$I_{BW} = \langle \langle I_R + I_G \rangle - I_B \rangle,$$

where the $\langle \rangle$ operation is the stretching function, I_{BW} is the image in black and white, and I_R , I_G , and I_B are the red, green, and blue channels. After that, a threshold value is used to eliminate spurious pixels in the background.

Deviations of the hand within the image, due mainly to small variations of the position of the camera, are corrected through resizing and rotation of the image [5], [12]. This is performed by locating two of the tops in the platforms (the one that the little finger presses and the top located between the forefinger and the middle finger). Then, an edge detection algorithm, based on the Sobel function, is applied to extract the contour of the hand.

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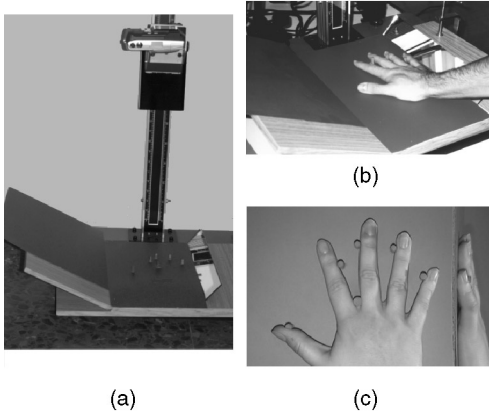


Fig. 1. Different views of the prototype designed: (a) Platform and camera, (b) placement of the user's hand, and (c) photograph taken.

2.3 Measurements

After preprocessing, the resulting image is a contour of the palm and the side view of the hand with the tops in a determined position. This simplifies the measurement algorithms and enables us to make all the measurements in pixels or angles. The main distances and angles measured are represented in Fig. 2a. These can be divided into four different categories:

1. **Widths:** Each of the four fingers is measured in different heights (mainly four of them), avoiding the pressure points near the tops (w_{11} - w_{44}). The width of the palm is also measured (w_0) and the distances among the three interfinger points (P_1 , P_2 , and P_3), in vertical and horizontal coordinates.
2. **Heights:** The middle finger, the little finger, and the palm (h_3 , h_2 , and h_1).
3. **Deviations:** Distance between a middle point of the finger and the middle point of the straight line between the interfinger point and the last height where the finger width is measured.

$$deviation = P_{12}^X - \left(\frac{P_{14}^X - P_1^X}{P_{14}^Y - P_1^Y} \right) (P_{12}^Y - P_1^Y),$$

where the subindices represent the measure point and the superindices represent the coordinate. Fig. 2b shows detail of how the deviations are computed.

4. **Angles:** between the interfinger points and the horizontal.

In order to minimize the variation of the distances measured with the weight gained or lost by the user, all distances are taken relative to a determined measure. The vertical coordinates, where the measurement points are located, are determined by the interfinger points and the tops, enabling us to profit from the whole length of the user's fingers and reducing errors in the measurement algorithm.

2.4 Feature Selection and Feature Vector Size

As seen in the previous section, 31 features have been extracted (21 widths, three heights, four deviations, and three angles). Once the features from a relative large number of users with several photographs obtained from each user, a statistical analysis has been performed to determine the discriminability of the features obtained. This is analyzed by a ratio F between the interclass and the intraclass variabilities. The higher this ratio, the more discriminant the feature is.

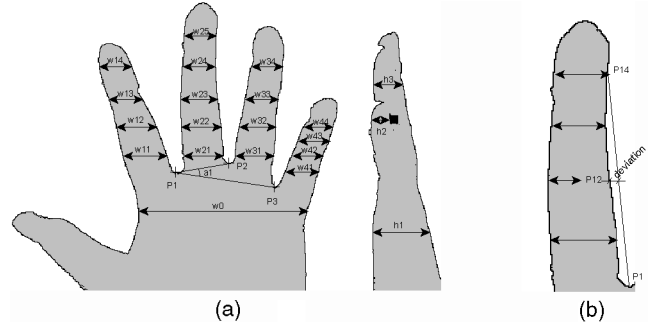


Fig. 2. (a) Location of measurement points for feature extraction. (b) Detail of the deviation measurement.

$$F_j = \frac{\text{interclass variability}}{\text{intraclass variability}} = \frac{\left(V \frac{1}{N} \sum_{i=1}^N \bar{f}_j^2 \right)}{\frac{1}{N} \sum_{i=1}^N V(f_j^i)},$$

where F_j is the ratio for the j th feature, V is the standard deviation function, N is the number of classes (users), f_j^i is the j th feature of the i th class and $\bar{f}_j$ is the mean of the j th features of the i th class. After this study, the number of discriminant features has been reduced to 25.

3 CLASSIFICATION AND VERIFICATION

The feature vectors obtained should enter a comparison process to determine the user whose hand photograph was taken. This comparison is to be made against user templates, which will be calculated depending on the comparison algorithm used. Therefore, in this section, the methods used will be explained, also analyzing the way the user template is calculated.

3.1 Euclidean Distance

The Euclidean distance ([1], [11]), considered the most common technique of all, performs its measurements with the following equation:

$$d = \sqrt{\sum_{i=1}^L (x_i - t_i)^2},$$

L being the dimension of the feature vector, x_i the i th component of the sample feature vector, and t_i the i th component of the template feature vector. The template vector dimension is, therefore, the same as one of the sample vectors. In order to calculate such a template, a set of photographs is taken and the mean of the resulting set of feature vectors is taken as the user's template.

3.2 Hamming Distance

This distance doesn't measure the difference between the components of the feature vectors, but the number of components that differ in value. As it is typical that all the components differ between samples of the same user, it is necessary to follow another approach for the template calculation different from the one used for the Euclidean distance. Based on the assumption that the feature components follow a Gaussian distribution, not only the mean of the set of initial samples is obtained, but also a factor of the standard deviation of the samples. Therefore, the template dimension will grow from 25 to 50 bytes. In the comparison process, the number of components of the feature vector falling outside the area defined by the template parameters is counted, obtaining the Hamming distance.

TABLE 1
Classification Success Compared to the Enrollment Set Size and the Feature Vector Dimension

		Euclidean	Hamming	GMMs	RBF
No. enrollment vectors (25 features)	3	86 %	75 %	88 %	90 %
	4	85 %	82 %	93 %	91 %
	5	86 %	87 %	96 %	91 %
Feature vector dimension (5 enrollment vectors)	25	86 %	87 %	96 %	91 %
	21	84 %	86 %	97 %	95 %
	15	86 %	88 %	96 %	89 %
	9	77 %	75 %	91 %	82 %

$$d(x_i, t_i^m) = \# \left\{ i \in \{1, \dots, L\} / |x_i - t_i^m| > t_i^v \right\},$$

where d is the Hamming distance, L is the dimension of the feature vectors, x_i is the i th component of the sample vector, t_i^m is the mean for the i th component, and t_i^v the factor of the standard deviation for the i th component.

3.3 Gaussian Mixture Models (GMMs)

This is a pattern recognition technique that uses an approach between the statistical methods and the neural networks. It is based on modeling the patterns with a determined number of Gaussian distributions, giving the probability of the sample belonging that class or not. Based on the works carried out by Reynolds and rose ([9], [10]), the probability density of a sample belonging to a class u is:

$$p(\vec{x}/u) = \sum_{i=1}^M \frac{c_i}{(2\pi)^{L/2} |\Sigma_i|^{1/2}} \exp \left\{ -\frac{1}{2} (\vec{x} - \vec{\mu}_i)^T \Sigma_i^{-1} (\vec{x} - \vec{\mu}_i) \right\},$$

c_i being the weights of each of the Gaussian models, μ_i the mean vector of each model, Σ_i the covariance matrix of each model, M the number of models, and L the dimension of feature vectors.

The GMM should be trained ([9], [10]), obtaining a GMM for each user, so the template of that user will be the final value of c_i , μ_i , Σ_i , and M , which greatly increases the template size.

3.4 Radial Basis Function Neural Networks (RBF)

The RBF is a two layer neural network ([2], [3], [6], and [11]), where the first layer is based on a radial basis function, such as a Gaussian distribution, and the second is a linear layer. To train the net, a set of feature vectors from all the users enrolled in the system should be used and each output will correspond to each class. If a verification system is considered, each user should have his own network which will have one single output. But, to train each user network, not only samples from this user should be applied but also a selection of other users, enabling the neural network to determine when the sample belongs to that user or not. This means that, in order to enable a biometric verification system based on RBF, a feature vector database should be stored for training new users which could be against some systems restrictions (confidentiality of the users' templates, etc.). For this reason, RBFs have been removed for the study of the verification system.

The user's template for this verification scheme is defined by the weights and bias of each of the neurons in the two layers of the RBF trained for that user. But, considering a classification scheme, the whole RBF neural network will include all users' templates.

4 EXPERIMENTAL RESULTS

This section reports some experimental results obtained by the authors at the E.T.S.I. de Telecomunicacion in the Madrid Polytechnic University. The work covered all the facts stated in the previous sections, detailing in this section the database used, the enrollment process, and final results in classification, i.e., in the Biometric Recognition scheme and in biometric verification.

4.1 Database

The experimentation started with the creation of a hand photograph database, due to the fact of not having a public one. The final database is composed of 10 photographs (taken on different days throughout three months) of 20 people of different ages, sex, profession, and living style. With this set of users, two main conclusions were obtained. First, all users, although most of them were not compromised with the experiment, showed great acceptance of the system, noting the ease of usage, needing only a few indications in their first capture. Second, the preprocessing algorithms were robust enough to allow colored skin users to use the system, as one of the users was Bengali.

4.2 Enrollment

As stated in Section 3, to calculate the user's template in the first three comparison methods used only needs a set of samples from that user, while the fourth method needs not only that set of samples, but a number of feature vectors from other users. One of the tasks to be studied for the enrollment process is the number of feature vectors that form the enrollment set that will be used to calculate the user's template. It is obvious that the bigger the number of samples used, the better the template calculated will be. However, the enrollment process means that several photographs of the user should be taken so, if the number of photographs is large enough, it could lead to rejection from the user. Therefore, a number between three and five samples for the enrollment set is considered sufficient.

4.3 Results in Classification

Two main analyses have been made to obtain results in biometric classification (i.e., biometric recognition). The first one performs the classification with the 25 initially selected features for different size of the enrollment set of feature vectors (3, 4, and 5), obtaining the results in Table 1. It can be seen that Euclidean distance and RBFs do not suffer apparent variation with the enrollment set size, while the other two methods show a great improvement. Also, with five vectors in the enrollment set, GMM shows the best results, achieving a 96 percent of success, while the metric methods are much below this number (about 87 percent), with a better performance of the neural network approach (91 percent).

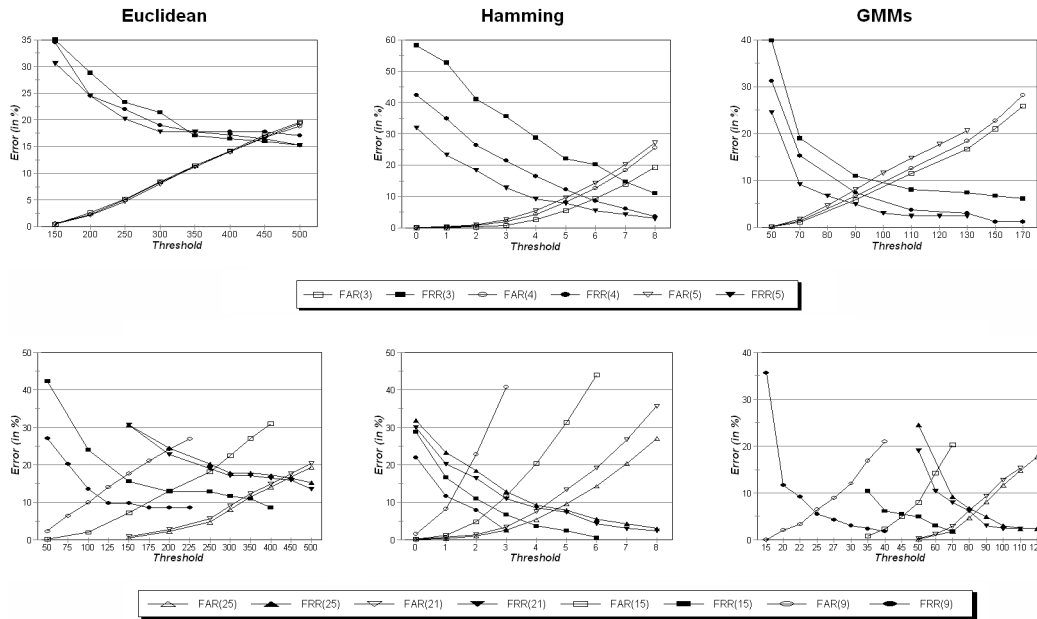


Fig. 3. Figures obtained. The upper ones refer to changes in enrollment set size. The bottom ones refer to changes in the feature vector dimension.

Once the enrollment set size is fixed to five, lower feature vector lengths have been studied: 21 (the feature vector without the deviations), 15 (with the deviations but without half of the finger widths), and nine (without deviations, half of the finger widths, and interfinger measurements). The results obtained can also be seen in Table 1, where the number of 15 features shows the best results for Euclidean and Hamming distances, while RBFs and GMMs achieve their best results with 21 features. A reduction to nine features leads to great loss in the success rate, causing us to consider such a reduction not useful.

4.4 Results in Verification

The same two analyses mentioned above were performed for biometric verification. The results obtained are given in False Acceptance Rate (FAR) and False Rejection Rate (FRR). Results obtained can be seen in Fig. 3, where it can be seen that, as stated in Section 3, the neural network approach is not present.

About the enrollment set size, the conclusions obtained are similar to the ones of the classification system (five vectors obtain the best results and Euclidean distance doesn't suffer any variation). Looking at the results about the feature vector size, three main results are obtained:

1. GMM still shows the best results.
2. The Equal Error Rate (where FAR = FRR), remains similar in each technique for the different feature vector sizes.
3. But, from the above point, the variation of FAR and FRR is more acute when the number of features is nine, being smoother when the number of features is 21 or 25. Results for 15 features are similar to the ones obtained for 21, being able to choose between both dimensions according to memory availability.

5 CONCLUSIONS

A hand geometry-based biometric identification system has been reported. A detailed explanation of the different blocks has been stated and different approaches have been studied, especially for the comparison block. Experimental results have been shown, giving a great performance of the system for medium security environments, achieving up to 97 percent of success in

classification and error rates much below 10 percent in verification. From the comparison methods, Gaussian Mixture Modeling (GMM) has been revealed as the one with the best performance through a higher computational cost and template size. This system as designed currently is considered a good alternative for security applications, although further work is being carried out to achieve better results and ease in implementation. Further work should be applied to increase the database size in order to study if the results obtained can be generalized.

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REFERENCES

- [1] R.O. Duda and P.E. Hart, *Pattern Classification and Scene Analysis*. John Wiley & Sons, 1973.
- [2] S. Haykin, *Neural Networks: A Comprehensive Foundation*. Prentice Hall, 1994.
- [3] D.R. Hush and B.G. Horne, "Progress in Supervised Neural Networks. What's New since Lippmann?" *IEEE Signal Processing Magazine*, pp. 8-39, 1993.
- [4] A.K. Jain, R. Bolle, and S. Pankanti, *BIOMETRICS Personal Identification in Networked Soc.*, p. 411, Kluwer Academic, 1999.
- [5] A.K. Jain, *Fundamentals of Digital Image Processing*. Prentice Hall, 1988.
- [6] R.P. Lippmann, "An Introduction to Computing with Neural Nets," *IEEE ASSP Magazine*, pp. 4-22, 1987.
- [7] B. Miller, "Vital Signs of Identity," *IEEE Spectrum*, pp. 22-30, Feb. 1994.
- [8] Recognition Systems Inc., web page: <http://www.recogsys.com>.
- [9] D.A. Reynolds and R.C. Rose, "Robust Text-Independent Speaker Identification Using Gaussian Mixture Speaker Models," *IEEE Trans. Speech and Audio Processing*, vol. 3, no. 1, pp. 72-83, 1995.
- [10] D.A. Reynolds, "Speaker Identification and Verification Using Gaussian Mixture Speaker Models," *Speech Comm.*, vol. 17, pp. 91-108, 1995.
- [11] J. Schürmann, *Pattern Classification. A Unified View of Statistical and Neural Approaches*. John Wiley & Sons, 1996.
- [12] R.J. Schalkoff, *Digital Processing and Computer Vision*. John Wiley & Sons, 1989.